
PRAXIS: Learning Dynamics of Self-Improving Models with Symbolic Archives

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Abstract

Self-improving learning systems couple adaptive data selection, optimization, and auxiliary symbolic components, leading to time-varying training objectives that fall outside the stationary assumptions of standard learning theory. This work introduces PRAXIS, a formal co-evolutionary framework that models generators, learners, and symbolic archives as interacting dynamical processes with explicitly regularized updates. A KL-constrained generator update is shown to induce a uniform, non-asymptotic bound on the per-iteration drift of the time-varying mixture objective over a finite horizon. For the symbolic archive, a high-probability analysis establishes exponential suppression of persistently non-competitive programs relative to an elite set under sub-Gaussian noise. These results characterize selection and suppression as structural properties of the update dynamics rather than heuristic design choices. Empirical analyses demonstrate that observed generator drift and archive concentration closely match the predicted theoretical bounds.

1. Introduction

Modern learning systems increasingly aim to improve autonomously by adapting not only model parameters but also the data, tasks, or auxiliary inductive signals that shape training. This paradigm appears in self-training (Wu et al., 2025; Behrouz et al., 2025), curriculum learning (Bengio et al., 2009; Shi et al., 2025; Wang et al., 2023), dataset distillation (Li et al., 2024; Howard et al., 2024; Li et al., 2021), active learning (Tamkin et al., 2022; Parvaneh et al., 2022), and recent work on self-evolving language models (Huang et al., 2025; Yu et al., 2025; Zhang et al., 2025). However, existing approaches treat such selection mechanisms as externally specified or only weakly adaptive. Data distri-

butions, supervision signals, and verification procedures are typically fixed in advance, or updated independently of the learner’s internal dynamics. Even label-free reinforcement learning relies on predefined task families or environment distributions (Mahan et al., 2024; Zou et al., 2021), while recent self-improvement methods for large language models (LLMs) depend on explicit executors, reward models, or domain-specific verifiers (Jin et al., 2025; Ma et al., 2025). In domains involving visual perception or symbolic reasoning where no reliable semantic oracle exists, these assumptions limit a system’s ability to regulate objective drift or adapt inductive structure.

Recent work has explored learning systems that combine multiple adaptive components, including adaptive task distributions and symbolic or heuristic modules (Solko-Breslin et al., 2024; Naik et al., 2024). While such approaches demonstrate the empirical promise of co-adaptive pipelines, they lack a unified formal model of the learning dynamics induced by interacting components, leaving questions of stability, objective drift, and long-horizon behavior largely unresolved. This paper introduces PRAXIS, a learning-theoretic framework that formalizes self-improving learning as a closed-loop dynamical process in which data selection, optimization, and symbolic inductive bias evolve jointly. By modeling the interaction between adaptive data distributions, time-varying objectives, and symbolic structure, PRAXIS enables principled analysis of stability and tracking in non-stationary learning environments.

The key contributions of this work are: i). A unified dynamical framework modeling joint evolution of data selection, learning objectives, and symbolic structure through explicit feedback. ii). Theoretical guarantees on controlled objective drift, exponential suppression of persistently non-informative symbolic programs, and stability of solver optimization under time-varying objectives. iii). An empirical instantiation on the MIT Indoor Scenes (Quattoni & Torralba, 2009) dataset demonstrating the predicted stability and curriculum dynamics in deep learning systems.

2. Preliminaries

Let $\mathbb{R}$ denote the real numbers, $\Delta(\mathcal{I})$ the probability simplex over a finite index set $\mathcal{I}$, and $\mathbb{E}[\cdot]$ expectation with respect to the distribution specified by context. Let $\mathcal{U}$ de-

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note a universe space indexing a family of supervised data-generating distributions. Each universe $u \in \mathcal{U}$ induces a distribution D_u over labeled examples $(x, y) \in \mathcal{X} \times \mathcal{Y}$, differing in properties such as data volume, bias, augmentation strength, or noise. The simplex $\Delta(\mathcal{U})$ denotes distributions over universes, and $\pi_0 \in \Delta(\mathcal{U})$ denotes a fixed reference prior. For $q, \pi \in \Delta(\mathcal{U})$, the Kullback–Leibler (KL) divergence is defined as $\text{KL}(q\|\pi) = \int_{\mathcal{U}} \log\left(\frac{dq}{d\pi}\right) dq$, whenever q is absolutely continuous with respect to π , and infinite otherwise.

A *Learner* is a parametric predictor $f_\theta : \mathcal{X} \rightarrow \Delta(\mathcal{Y})$, $\theta \in \Theta$, mapping inputs to predictive label distributions. The predictive output $z_\theta(x) := f_\theta(x)$ constitutes the sole interface exposed to symbolic components; all symbolic interaction operates directly on the Learner’s predictive distribution. A *Generator* is a stochastic policy $q_\phi \in \Delta(\mathcal{U})$, $\phi \in \Phi$, which specifies a sampling distribution over universes and thereby determines the mixture distribution from which training data are drawn. The framework maintains a finite *archive* of symbolic programs $\mathcal{P} = \{p_1, \dots, p_m\}$. Each program $p_j : \Delta(\mathcal{Y}) \rightarrow \mathcal{Y}$ is a deterministic rule mapping the Learner’s predictive distribution to a hard label. Programs encode heuristic or structural biases and are not assumed to be semantically correct. Each program p_j is associated with a score $s_j \in \mathbb{R}$, inducing a Gibbs distribution over the archive: $w_j = \frac{\exp(\beta s_j)}{\sum_{k=1}^m \exp(\beta s_k)}$, $\beta > 0$, where $w \in \Delta(\mathcal{P})$ represents a soft inductive bias over symbolic hypotheses.

3. Problem Formulation

Learning is defined over a universe space $\mathcal{U}$ indexing a family of supervised data-generating distributions $\{D_u\}_{u \in \mathcal{U}}$. For a Learner $f_\theta : \mathcal{X} \rightarrow \Delta(\mathcal{Y})$, the population loss on universe u is $L(\theta, u) = \mathbb{E}_{(x,y) \sim D_u} [\ell(f_\theta(x), y)]$, where ℓ is a bounded supervised loss. Given a universe distribution $q \in \Delta(\mathcal{U})$, the induced mixture risk is $\mathcal{R}(\theta, q) = \mathbb{E}_{u \sim q} L(\theta, u)$.

The Generator adapts the universe distribution to emphasize learning-relevant regions of $\mathcal{U}$. Deviations from a reference prior $\pi_0 \in \Delta(\mathcal{U})$ are regularized via a KL penalty, yielding the variational objective

$$\min_{\theta \in \Theta} \max_{q \in \Delta(\mathcal{U})} \left\{ \mathbb{E}_{u \sim q} L(\theta, u) - \frac{1}{\eta} \text{KL}(q\|\pi_0) \right\}, \quad (1)$$

where $\eta > 0$ controls the adaptation rate. In practice, the maximization over q is implemented through KL–proximal updates, ensuring smooth generator dynamics and bounded objective drift.

The symbolic archive $\mathcal{P} = \{p_j\}_{j=1}^m$ introduces auxiliary inductive structure through a nonnegative consistency functional. For a labeled example (x, y) , the archive-induced penalty is defined as $\Psi(f_\theta, \mathcal{P}; x, y) =$

$\sum_{j=1}^m w_j b_j(f_\theta; x, y) \rho_j(f_\theta; x, y)$, where w_j are the current archive weights, and the discrepancy and blame terms selectively suppress symbolic influence that conflicts with reliable neural predictions. This yields the augmented per-universe loss

$$L_{\text{aug}}(\theta, u) = \mathbb{E}_{(x,y) \sim D_u} [\ell(f_\theta(x), y) + \lambda \Psi(f_\theta, \mathcal{P}; x, y)], \quad (2)$$

with $\lambda > 0$ controlling the strength of symbolic regularization.

Combining adaptive universe selection with archive-augmented learning yields the regularized min–max objective

$$\min_{\theta \in \Theta} \max_{q \in \Delta(\mathcal{U})} \left\{ \mathbb{E}_{u \sim q} L_{\text{aug}}(\theta, u) - \frac{1}{\eta} \text{KL}(q\|\pi_0) \right\}. \quad (3)$$

This objective is not solved to equilibrium; instead, it induces a coupled closed-loop dynamics in which the Learner and Generator evolve jointly. The Archive is not optimized directly, and influences learning only through the time-varying weights w_j appearing in Ψ .

4. Solution Approach

PRAXIS realizes the variational formulation of Section 3 as a discrete-time stochastic process. At iteration $t = 0, 1, 2, \dots$, the system maintains a coupled state (θ_t, s_t, ϕ_t) , where $\theta_t \in \Theta$ are Learner parameters, $s_t = (s_{1,t}, \dots, s_{m,t})$ are Archive scores, and $\phi_t \in \Phi$ parameterizes a Generator distribution $q_t := \pi_{\phi_t} \in \Delta(\mathcal{U})$. The induced Archive weights are $w_{j,t} = \exp(\beta s_{j,t}) / \sum_k \exp(\beta s_{k,t})$.

Updates are performed online and are not solved to equilibrium. Instead, each component is updated using stochastic first-order information computed under the current state of the others. All coupling between components occurs through the augmented loss L_{aug} and the current weights (q_t, w_t) , yielding a closed-loop learning process in which data selection, inductive structure, and model parameters evolve jointly.

4.1. Sampling and Estimation

At iteration t , the Generator distribution q_t is used to sample K universes $u_t^{(1)}, \dots, u_t^{(K)} \sim q_t$ independently. From each sampled universe $u_t^{(k)}$, an empirical dataset $\mathcal{D}_t^{(k)}$ is drawn from $D_{u_t^{(k)}}$. These samples define a Monte Carlo approximation to expectations with respect to the mixture distribution induced by q_t . Let $\mathcal{D}_t := \bigcup_{k=1}^K \mathcal{D}_t^{(k)}$ denote the aggregated dataset at iteration t .

This sampling mechanism provides a unified source of signal for the Learner, Archive, and Generator updates, en-

ensuring that all components respond to the same underlying universe mixture at each iteration.

4.2. Learner Update

At each iteration t , the Learner produces predictive distributions $z_{\theta_t}(x) = f_{\theta_t}(x) \in \Delta(\mathcal{Y})$. Symbolic programs in the archive operate exclusively on these predictive distributions, and no additional latent representations are assumed. This design enforces a strict interface between neural and symbolic components. For any universe $u \in \mathcal{U}$, the augmented population loss is

$$L_{\text{aug}}(\theta_t, u) = \mathbb{E}_{(x,y) \sim D_u} \left[\ell(f_{\theta_t}(x), y) + \lambda \Psi(f_{\theta_t}, \mathcal{P}_t; x, y) \right], \quad (4)$$

where ℓ is a bounded supervised loss and Ψ is the Archive-induced consistency functional. By construction of the consistency functional (Section 3), $0 \leq \Psi \leq 1$, and therefore

$$0 \leq L_{\text{aug}}(\theta, u) \leq L_{\text{max}} + \lambda =: M \quad \forall \theta, u.$$

Given q_t , the induced mixture objective is $F_t(\theta) = \mathbb{E}_{u \sim q_t} L_{\text{aug}}(\theta, u)$. The Learner update is implemented via a stochastic descent step $\theta_{t+1} = \theta_t - \alpha_t g_t$, where g_t is an unbiased estimator of $\nabla F_t(\theta_t)$ computed from minibatches drawn from $\mathcal{D}_t$. Gradients propagate only through $f_{\theta_t}(x)$; symbolic programs are treated as fixed, non-differentiable mappings and influence the Learner exclusively through the weighted consistency functional Ψ , with each program's contribution scaled by its weight $w_{j,t}$.

4.3. Archive Update

At each iteration t , the framework maintains a finite archive of symbolic programs $\mathcal{P} = \{p_1, \dots, p_m\}$. Each program $p_j : \Delta(\mathcal{Y}) \rightarrow \mathcal{Y}$ is a deterministic mapping from the Learner's predictive distribution to a hard label. Programs encode heuristic or structural inductive biases and are not assumed to be semantically correct. The archive may be augmented at discrete expansion events; between such events, $\mathcal{P}$ is treated as fixed, and learning dynamics act only on the associated scores and weights.

Each program p_j is assigned a scalar utility $U_{j,t} = F_{j,t} - \gamma B_{j,t}$, where $\gamma > 0$ controls the penalty assigned to counterproductive symbolic influence. The fitness term

$$F_{j,t} = \frac{1}{|\mathcal{D}_t|} \sum_{(x,y) \in \mathcal{D}_t} \mathbf{1}[p_j(f_{\theta_t}(x)) = y]$$

measures empirical agreement with ground-truth labels under the current universe mixture, while the blame term

$$B_{j,t} = \frac{1}{|\mathcal{D}_t|} \sum_{(x,y) \in \mathcal{D}_t} \mathbf{1}[f_{\theta_t}(x) = y \wedge p_j(f_{\theta_t}(x)) \neq y]$$

penalizes symbolic disagreement only on samples already classified correctly by the Learner, thereby gating blame by learner reliability.

The empirical utility used in the update is $\hat{U}_{j,t} = U_{j,t} + \xi_{j,t}$, where $\xi_{j,t}$ captures stochastic estimation noise due to finite sampling. Archive scores evolve according to $s_{j,t+1} = s_{j,t} + \eta_p \hat{U}_{j,t}$, where $\eta_p > 0$ controls the rate at which empirical utility is accumulated over time. Scores are mapped to archive weights via a Gibbs distribution $w_{j,t} = \frac{\exp(\beta s_{j,t})}{\sum_{k=1}^m \exp(\beta s_{k,t})}$, $\beta > 0$.

This update induces a competitive selection process in which symbolic programs accumulate influence in proportion to sustained empirical utility, while programs that are persistently uninformative or contradictory are progressively downweighted.

4.4. Generator Update

The Generator maintains a parametric distribution $q_t = \pi_{\phi_t} \in \Delta(\mathcal{U})$ over the universe space and adapts it in response to the evolving Learner-Archive state. At iteration t , universes $u_t^{(1)}, \dots, u_t^{(K)} \sim q_t$ are sampled and used to update the Learner. After the Learner update θ_{t+1} , each sampled universe is assigned a scalar reward $R_t^{(k)} = \hat{L}_{\text{aug}}(\theta_{t+1}, u_t^{(k)}) + \epsilon_t^{(k)}$, where $\hat{L}_{\text{aug}}$ is the empirical augmented loss evaluated post-update and $\epsilon_t^{(k)}$ is an exploration noise term. Post-update evaluation biases the Generator toward universes that remain informative under adaptation.

The Generator update minimizes the KL-regularized objective

$$\begin{aligned} \mathcal{J}_t(\phi) = & -\frac{1}{K} \sum_{k=1}^K (R_t^{(k)} - b_t) \log \pi_{\phi}(u_t^{(k)}) \\ & + \frac{1}{\eta_1} \text{KL}(\pi_{\phi} \| \pi_{\phi_t}) + \frac{1}{\eta_2} \text{KL}(\pi_{\phi} \| \pi_0), \end{aligned} \quad (5)$$

where b_t is a variance-reduction baseline and π_0 is a fixed reference prior. The first KL term enforces a trust-region constraint limiting step-to-step distributional movement, while the second prevents collapse onto degenerate subsets of $\mathcal{U}$.

Updates are accepted only if the resulting distribution $q_{t+1} = \pi_{\phi_{t+1}}$ satisfies $\text{KL}(q_{t+1} \| q_t) \leq \kappa_t$, for a prescribed radius $\kappa_t > 0$. Generator parameters are updated via $\phi_{t+1} = \phi_t - \eta_{\phi} \nabla_{\phi} \mathcal{J}_t(\phi_t)$, which corresponds to a natural-gradient step on $\Delta(\mathcal{U})$ under a softmax parameterization. In expectation, these dynamics approximate ascent on the KL-regularized inner maximization of Eq. (3).

Collectively, these updates implement a stochastic primal-dual tracking procedure for the regularized min-max

objective in Eq. (3), with the Learner performing descent in θ and the Generator performing regularized ascent in q .

5. Theoretical Analysis of PRAXIS Dynamics

This section analyzes the stability properties of the learning dynamics induced by PRAXIS. Rather than establishing convergence to a stationary optimum, the analysis focuses on structural guarantees that align directly with the implemented update rules. Specifically, KL-regularized Generator updates are shown to control nonstationarity of the Learner’s population objective (Subsection 5.1), Archive weights are shown to suppress persistently uninformative symbolic programs under stochastic utility estimation (Subsection 5.2), and the Learner is shown to track the resulting time-varying objective under standard stochastic gradient assumptions (Subsection 5.3). While interpretations of the theory are provided here, the proofs are provided in the Appendix.

5.1. Generator-induced objective drift

Definition 5.1 (Time-Varying Mixture Objective). At iteration t , let $q_t := \pi_{\phi_t} \in \Delta(\mathcal{U})$ denote the Generator distribution. The associated time-varying mixture objective is $F_t(\theta) := \mathbb{E}_{u \sim q_t} L_{\text{aug}}(\theta, u)$, $\theta \in \Theta$, where $L_{\text{aug}}(\theta, u)$ is the augmented per-universe loss defined in Section 3.

Definition 5.2 (Uniform Objective Drift). The one-step uniform objective drift at iteration t is defined as $\delta_t := \sup_{\theta \in \Theta} |F_{t+1}(\theta) - F_t(\theta)|$.

Remark 5.3. The PRAXIS Generator update enforces a KL trust region, ensuring that consecutive universe distributions do not change arbitrarily between iterations. This constraint is the primary mechanism used to control objective drift.

Theorem 5.4 (KL-controlled drift of the time-varying mixture objective). *Let $\{q_t\}_{t \geq 0} \subset \Delta(\mathcal{U})$ be the Generator sequence and $F_t(\theta) = \mathbb{E}_{u \sim q_t} L_{\text{aug}}(\theta, u)$ the induced mixture objective. If the augmented loss satisfies the uniform bound $\sup_{\theta \in \Theta, u \in \mathcal{U}} |L_{\text{aug}}(\theta, u)| \leq M$, then, for every $t \geq 0$, $\delta_t = \sup_{\theta \in \Theta} |F_{t+1}(\theta) - F_t(\theta)| \leq M \|q_{t+1} - q_t\|_1 \leq M \sqrt{2 \text{KL}(q_{t+1} \| q_t)}$, whenever $\text{KL}(q_{t+1} \| q_t) < \infty$. Moreover, if the Generator updates satisfy a summable trust-region condition $\sum_{t=0}^{\infty} \text{KL}(q_{t+1} \| q_t) < \infty$, then $\text{KL}(q_{t+1} \| q_t) \rightarrow 0$ and hence $\delta_t \rightarrow 0$ as $t \rightarrow \infty$.*

Interpretation. Theorem 5.4 shows that Generator movement directly controls the nonstationarity of the Learner’s objective. As successive Generator updates become small in KL divergence, the learning problem becomes asymptotically stationary, even though no equilibrium is ever explicitly solved.

While δ_t bounds worst-case objective drift, learning proceeds along a specific stochastic trajectory. The next result

relates this uniform notion to the drift actually experienced along the Learner iterates.

Definition 5.5 (Along-iterate objective drift (magnitude)). Let $\{\theta_t\}_{t \geq 0}$ denote the (possibly stochastic) sequence of Learner iterates and $\{F_t\}_{t \geq 0}$ the associated mixture objectives. The one-step objective drift along the iterates is defined as $\Delta_t := \mathbb{E}[|F_{t+1}(\theta_{t+1}) - F_t(\theta_{t+1})|]$, where the expectation is taken over all algorithmic randomness up to iteration $t + 1$.

Lemma 5.6 (Along-iterate drift is dominated by uniform drift). *For all $t \geq 0$, $\Delta_t \leq \sup_{\theta \in \Theta} |F_{t+1}(\theta) - F_t(\theta)| = \delta_t$.*

Lemma 5.6 states that the expected objective change experienced by the Learner along its realized trajectory is bounded by the worst-case (uniform) objective drift. This permits uniform drift bounds to be used as conservative surrogates in tracking analyses. Under stronger assumptions on Generator stabilization, the objective drift decays at a quantifiable rate as shown below.

Definition 5.7 (Exponential Generator stabilization). The Generator is said to stabilize exponentially after T_0 if there exist constants $c > 0$, $\kappa > 0$, and $T_0 \in \mathbb{N}$ such that $\|q_{t+1} - q_t\|_1 \leq c e^{-\kappa(t-T_0)}$, $\forall t \geq T_0$.

Corollary 5.8 (Exponential Generator stabilization implies exponential objective stabilization). *Under the assumptions of Theorem 5.4, if the Generator stabilizes exponentially in the sense of Definition 5.7, then for all $t \geq T_0$, $\delta_t \leq M \|q_{t+1} - q_t\|_1 \leq M c e^{-\kappa(t-T_0)}$, and the along-iterate drift satisfies $\Delta_t \leq \delta_t \leq M c e^{-\kappa(t-T_0)}$.*

Remark (stabilization versus concentration). In finite universe spaces, measures such as entropy or the maximum mass $\max_u q_t(u)$ are often used to diagnose Generator concentration. Such concentration is compatible with stabilization measured in the ℓ_1 distance $\|q_{t+1} - q_t\|_1$ (equivalently, total variation distance) or in KL divergence: the distribution may become increasingly peaked while the one-step change between successive Generator iterates decays. For this reason, stabilization in KL geometry (or ℓ_1 distance), rather than concentration alone, is the appropriate condition governing the drift of the time-varying mixture objective.

5.2. Archive suppression under persistent utility advantage

Definition 5.9 (Population and empirical archive utilities). For each program p_j at iteration t , its population utility is $U_{j,t} := F_{j,t} - \gamma B_{j,t}$, where $F_{j,t}$ and $B_{j,t}$ denote symbolic fitness and blame, respectively, as defined in Section 4. The empirical utility used to update the archive is $\hat{U}_{j,t} = U_{j,t} + \xi_{j,t}$, where $\xi_{j,t}$ represents stochastic estimation noise arising from finite sampling over universes and data.

Definition 5.10 (Sub-Gaussian pairwise utility estimation

noise). Let $\{\mathcal{F}_t\}_{t \geq 0}$ denote the natural filtration generated by the PRAXIS dynamics. The noise process $\{\xi_{j,t}\}$ satisfies, for constants $\mu \geq 0$ and $\sigma > 0$: (i) Bounded conditional bias: For all j, t , $|\mathbb{E}[\xi_{j,t} | \mathcal{F}_{t-1}]| \leq \mu$ a.s. (ii) Pairwise sub-Gaussian increments: For all ordered pairs $(i, j) \in [m] \times [m]$, all t , and all $\lambda \in \mathbb{R}$, $\mathbb{E}\left[\exp\left(\lambda\left((\xi_{j,t} - \xi_{i,t}) - \mathbb{E}[\xi_{j,t} - \xi_{i,t} | \mathcal{F}_{t-1}]\right)\right) \middle| \mathcal{F}_{t-1}\right] \leq \exp\left(\frac{\lambda^2(2\sigma)^2}{2}\right)$ a.s.

Lemma 5.11 (Learner-coupled lower bound on utility gaps). Assume $0 \leq F_{k,t} \leq 1$ for all k, t , and that blame is learner-gated in the sense that blame is incurred only on samples correctly classified by the learner. Let $e_t := \mathbb{P}(\hat{y}_t(x) \neq y)$ denote the learner population error at iteration t . Then, for all k, t , $0 \leq B_{k,t} \leq 1 - e_t$. Consequently, for any programs i, j , the population utility gap satisfies $U_{i,t} - U_{j,t} \geq (F_{i,t} - F_{j,t}) - \gamma(1 - e_t)$. Define the endogenous margin lower bound $\varepsilon_{i,j,t}^{\text{endo}} := (F_{i,t} - F_{j,t}) - \gamma(1 - e_t)$. Then $U_{i,t} - U_{j,t} \geq \varepsilon_{i,j,t}^{\text{endo}}$ for all t .

Interpretation. Lemma 5.11 shows that as the learner improves ($e_t \downarrow$), blame becomes increasingly selective, enlarging utility gaps between programs. Archive suppression is therefore endogenously accelerated by learner progress rather than imposed externally.

Theorem 5.12 (High-probability archive weight suppression). Let archive scores evolve as $s_{j,t+1} = s_{j,t} + \eta_p \hat{U}_{j,t}$, $w_{j,t} = \frac{e^{\beta s_{j,t}}}{\sum_{k=1}^m e^{\beta s_{k,t}}}$, with $\beta > 0$, $\eta_p > 0$, and $\hat{U}_{j,t} = U_{j,t} + \xi_{j,t}$. Assume the noise process satisfies Definition 5.10. Fix a nonempty subset $\mathcal{S} \subseteq [m]$. For $j \notin \mathcal{S}$, define the utility gap to the elite set $\varepsilon_{j,t}^{\mathcal{S}} := \max_{i \in \mathcal{S}} (U_{i,t} - U_{j,t})$, $\varepsilon_t^{\mathcal{S}} := \min_{j \notin \mathcal{S}} \varepsilon_{j,t}^{\mathcal{S}}$. Then for any horizon $T \geq T_0$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the following holds uniformly for all $t \in \{T_0, \dots, T\}$ and all $j \notin \mathcal{S}$: $\frac{w_{j,t}}{\sum_{i \in \mathcal{S}} w_{i,t}} \leq \frac{w_{j,T_0}}{\sum_{i \in \mathcal{S}} w_{i,T_0}} \exp\left(-\beta \eta_p \sum_{\tau=T_0}^{t-1} \varepsilon_{j,\tau}^{\mathcal{S}} + \beta \eta_p \Gamma_{\text{subG}}(t, \delta)\right)$, where $\Gamma_{\text{subG}}(t, \delta) = 2\mu(t - T_0) + 2\sigma \sqrt{2(t - T_0) \log\left(\frac{m^2(T - T_0 + 1)}{\delta}\right)}$.

Interpretation. Theorem 5.12 shows that the archive update induces a principled selection mechanism: symbolic programs that are persistently less useful than an elite subset experience exponential decay in relative weight, up to stochastic fluctuations induced by finite sampling. The result holds with high probability under mild assumptions on utility estimation noise and does not require the learner to be optimal or converged. Crucially, suppression depends on cumulative utility gaps rather than instantaneous performance, ensuring robustness to transient noise and short-term reversals. As a consequence, the archive dynamics favor sustained empirical usefulness over momentary agreement, yielding a stable pruning effect even in a nonstationary learn-

ing environment.

Corollary 5.13 (Exponential suppression under persistent positive margins). Under the conditions of Theorem 5.12, suppose there exist constants $\varepsilon > 0$ and $T_1 \geq T_0$ such that $\sum_{\tau=T_1}^{t-1} \max(\varepsilon_{j,\tau}^{\mathcal{S}}, 0) \geq \varepsilon(t - T_1) \quad \forall t \leq T$. Then, with probability at least $1 - \delta$, the archive weights satisfy $\log \frac{w_{j,t}}{\sum_{i \in \mathcal{S}} w_{i,t}} \leq C - \beta \eta_p \varepsilon(t - T_1) + \beta \eta_p \Gamma_{\text{subG}}(t, \delta)$, $j \notin \mathcal{S}$, for a constant C depending only on the initial weights at T_1 . In particular, if $\mu = 0$, the exponent grows linearly in t , yielding exponential suppression up to sublinear stochastic fluctuations.

Corollary 5.13 strengthens Theorem 5.12 by identifying a simple sufficient condition under which archive suppression becomes exponential. If non-elite programs exhibit a persistent positive utility gap relative to an elite subset, then their relative weights decay exponentially fast, up to sublinear stochastic fluctuations induced by finite-sample noise. In this regime, archive selection behaves analogously to a replicator dynamic with endogenous margins: once a program is consistently dominated, its influence vanishes at a rate determined by the learning rate, inverse temperature, and the magnitude of the persistent utility gap. This formalizes how long-term symbolic competition induces effective archive compression without requiring explicit pruning heuristics.

5.3. Learner tracking under Generator-induced objective drift

Theorem 5.14 (SGD tracking under Generator-induced objective drift). Let $\{q_t\}_{t \geq 0}$ denote the Generator sequence and define the time-varying mixture objective $F_t(\theta) := \mathbb{E}_{u \sim q_t} L_{\text{aug}}(\theta, u)$, where $L_{\text{aug}}(\theta, u)$ is given in (2). Assume that each F_t is bounded below on Θ , i.e., there exists $F_{\min} \in \mathbb{R}$ such that $F_t(\theta) \geq F_{\min}$ for all $t \geq 0$ and $\theta \in \Theta$. Let the Learner update be $\theta_{t+1} = \theta_t - \eta g_t$, and let $\{\mathcal{F}_t\}_{t \geq 0}$ denote the natural filtration generated by all algorithmic randomness up to iteration t . Assume that for all $t \geq 0$ the stochastic gradient g_t is $\mathcal{F}_t$ -measurable and satisfies:

- (i) Unbiasedness: $\mathbb{E}[g_t | \mathcal{F}_{t-1}] = \nabla F_t(\theta_t)$.
- (ii) Bounded conditional variance: There exists $\sigma^2 > 0$ such that $\mathbb{E}[\|g_t - \nabla F_t(\theta_t)\|^2 | \mathcal{F}_{t-1}] \leq \sigma^2$.

Assume further that each F_t is L -smooth on Θ , i.e.,

$$\|\nabla F_t(\theta) - \nabla F_t(\theta')\| \leq L\|\theta - \theta'\|, \quad \forall \theta, \theta' \in \Theta.$$

Then, for any horizon $T \geq 1$ and stepsize $\eta \in (0, 1/L]$, $\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla F_t(\theta_t)\|^2] \leq \frac{2}{\eta T} (F_0(\theta_0) - F_{\min}) + \eta L \sigma^2 + \frac{2}{T} \sum_{t=0}^{T-1} \delta_t$, where the one-step uniform objective drift is defined as $\delta_t := \sup_{\theta \in \Theta} |F_{t+1}(\theta) - F_t(\theta)|$.

Interpretation. Theorem 5.14 bounds the average stationarity violation $\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla F_t(\theta_t)\|^2$ of stochastic gradient descent when optimizing a time-varying mixture objective. The first two terms coincide with standard non-convex SGD guarantees for a fixed objective, capturing optimization error due to initialization and gradient noise. The additional term $\frac{2}{T} \sum_{t=0}^{T-1} \delta_t$ explicitly quantifies the effect of Generator-induced nonstationarity, measuring how rapidly the underlying objective itself changes across iterations. Consequently, when the Generator stabilizes so that the cumulative drift $\sum_{t=0}^{T-1} \delta_t$ grows sublinearly in T (as established in Section 5.1), the Learner provably tracks a sequence of near-stationary points of $\{F_t\}$. In this regime, the Learner adapts fast enough to follow the evolving objective, despite continual changes in the universe distribution.

6. PRAXIS Experimental Setup

Dataset: Experiments are conducted on a subset of the MIT Indoor Scenes dataset. A 12-class subset is used to balance visual diversity with controlled evaluation, yielding approximately 1,200 images. Data are split into fixed 70/30 training and test partitions and per-class image counts are reported in Appendix C.

Universes and evaluation protocol: Training environments are modeled as a finite discrete universe space $\mathcal{U}$, where each universe corresponds to a specific combination of data subsampling size, augmentation strength, corruption type (none, blur, JPEG compression, or additive noise), and corruption severity. The Cartesian product of these factors yields a collection of several hundred distinct training universes. A Generator maintains a categorical distribution over $\mathcal{U}$ and selects universes for training over 50 iterations. Robustness is evaluated over a separate set of held-out universes constructed from parameter configurations disjoint from those used during training. Results are reported over 27 held-out universes, which are never sampled by the Generator and are used exclusively for evaluation.

Learner: The Learner is instantiated as a ResNet-34 convolutional neural network pretrained on ImageNet. Optimization is performed using Adam with a learning rate of 10^{-3} , selected via a preliminary Optuna-based hyperparameter search. Further implementation details are provided in Appendix D

Symbolic archive: PRAXIS maintains a finite archive of deterministic symbolic programs (see Appendix E), each mapping the Learner’s predictive distribution to a hard class label. Programs are manually curated to span multiple families of simple structural heuristics, including ranking-based rules, threshold-based rules, symmetry-breaking biases, temporal or modular patterns, and distributional preferences. Archive programs are assigned real-valued scores using the

fitness-blame mechanism described earlier. The archive influences learning only through its weighted regularization term and no explicit pruning or manual selection is performed.

LLM-assisted archive expansion: To expand the symbolic hypothesis space, additional archive programs are periodically proposed by a large language model. The GPT-4o LLM is used to propose programs once every two iterations from a fixed library of predefined templates with bounded parameters. The language model does not generate executable code, observe training data, gradients, or model parameters. All proposed programs are initialized with neutral scores and are subject to the same archive update dynamics as manually specified programs. Prompting details and template definitions are provided in Appendix E.

7. Results

Empirical evaluation is organized around three research questions (RQs) that directly correspond to the theoretical properties established in the preceding sections. The findings for each RQ are reported below.

RQ1: Does PRAXIS, which couples adaptive universe selection with symbolic archive regularization, improve robustness compared to existing adaptive and non-adaptive baselines?

Baselines: PRAXIS is compared against four representative baselines. *Empirical Risk Minimization (ERM)* trains on a fixed uniform mixture of universes without adaptation. *Group Distributionally Robust Optimization (Group DRO)* optimizes worst-group performance over universes via adaptive reweighting. *Curriculum Learning (CL)* adaptively reshapes the training distribution over universes without mixture regularization or symbolic structure. *Bandit-based sampling* treats universes as arms in a stochastic multi-armed bandit and adapts selection based on short-term feedback. All methods share the same Learner architecture, and evaluation protocol as PRAXIS.

Table 1 reports per-universe results, and evaluation metrics for each of the different techniques. PRAXIS achieves the highest mean accuracy (0.694) and the strongest tail robustness ($\text{CVaR}_{10} = 0.639$) among all evaluated methods. Relative to ERM, PRAXIS improves mean accuracy by 1.6 percentage points (0.694 vs. 0.678) while maintaining comparable cross-universe stability (standard deviation 0.031 vs. 0.029). Compared to adaptive baselines, PRAXIS exhibits substantially stronger robustness on difficult universes. Its CVaR_{10} exceeds that of Group DRO by 9.5 percentage points (0.639 vs. 0.541), curriculum learning by 11.4 points (0.639 vs. 0.522), and bandit-based sampling by 21.1 points (0.639 vs. 0.428). PRAXIS also shows markedly lower variance across universes than all adaptive baselines, indi-

Table 1. Per-universe evaluation on a fixed test-universe set

Method	Mean	Min	CVaR ₁₀	Std
ERM	0.678	0.617	0.630	0.029
Group DRO	0.636	0.544	0.541	0.049
CL	0.615	0.484	0.522	0.059
Bandit	0.605	0.375	0.428	0.084
PRAXIS (Ours)	0.694	0.602	0.639	0.031

For each trained learner θ and each test universe $u \in \mathcal{U}_{\text{test}}$, the per-universe accuracy is $\text{Acc}(\theta; u)$. **Mean** $:= \frac{1}{|\mathcal{U}_{\text{test}}|} \sum_u \text{Acc}(\theta; u)$.

Min $:= \min_u \text{Acc}(\theta; u)$ (worst-case accuracy). **CVaR₁₀** $:= \frac{1}{k} \sum_{i=1}^k \text{Acc}_{(i)}$ where $\text{Acc}_{(1)} \leq \dots \leq \text{Acc}_{(|\mathcal{U}_{\text{test}}|)}$ are sorted accuracies and $k = \lceil 0.10 |\mathcal{U}_{\text{test}}| \rceil$ (mean of the bottom 10% universes).

Std $:= \sqrt{\frac{1}{|\mathcal{U}_{\text{test}}|} \sum_u (\text{Acc}(\theta; u) - \text{Mean})^2}$ (standard deviation).

cating reduced performance volatility under distributional shift. While ERM attains the highest single worst-case accuracy, robustness in this work is defined by tail behavior across many adverse universes, as captured by CVaR₁₀ and cross-universe variance. Under these criteria, PRAXIS consistently outperforms both static and adaptive baselines. Overall, these results demonstrate that PRAXIS achieves a favorable balance between average accuracy, tail robustness, and stability across heterogeneous training environments.

RQ2: Are the empirical robustness gains of PRAXIS explained by stable generator dynamics consistent with the KL-controlled objective drift analysis?

Generator-Induced Drift: Figure 1 reports the evolution of the drift proxy $2\text{KL}(q_{t+1} \| q_t)$, which upper-bounds the per-iteration change in the mixture objective $F_t(\theta)$ up to the bounded-loss constant (Theorem 5.4). After an initial transient phase (first ten iterations), PRAXIS exhibits consistently small generator updates, with a maximum drift proxy of 0.1133 and 95th/99th percentiles of 0.0888 and 0.1069, respectively.

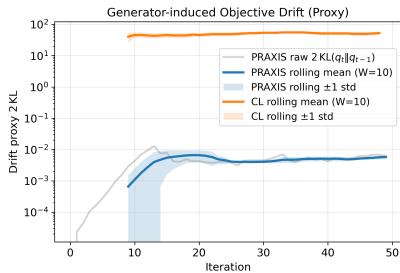


Figure 1. Generator-induced drift proxy (log scale). Rolling mean of $2\text{KL}(q_{t+1} \| q_t)$ across iterations for PRAXIS and CL. Reported summary statistics exclude the first ten iterations.

For comparison, CL, the closest adaptive baseline, exhibits drift proxy values of 7.4803 at the maximum and corresponding percentiles, exceeding PRAXIS by nearly two

orders (≈ 1.83) of magnitude. Non-adaptive baselines are omitted as their training distributions are fixed by construction. These results place PRAXIS in a distinct adaptation regime characterized by bounded, incremental distributional updates induced by explicit KL regularization.

Generator Dynamics: Figure 2 characterizes internal generator behavior. Entropy $H(q_t)$ decreases smoothly from 4.6791 to 4.2689, while the maximum universe probability increases from 0.0110 to 0.0719, indicating gradual concentration onto a subset of informative universes. Despite this specialization, successive KL updates remain small throughout training, with mean $\text{KL}(q_{t+1} \| q_t)$ increasing from 3.27×10^{-4} to 2.69×10^{-3} and a maximum observed value of 6.42×10^{-3} . This confirms that concentration arises through many controlled reweighting steps rather than abrupt collapse or oscillation.

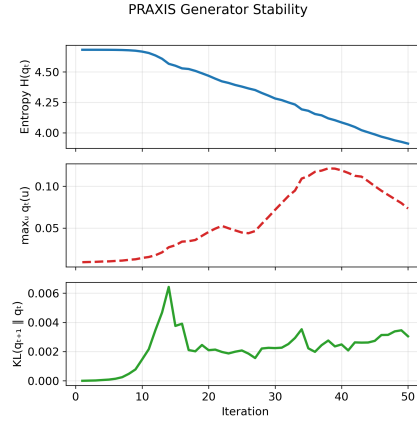


Figure 2. Evolution of Generator Probability Mass

Learner Stability: Figure 3 examines learner behavior under the adaptive training distributions induced by the PRAXIS generator. Over training, the rolling mean of the cross-entropy loss decreases from 0.1504 to 0.1066, while the rolling standard deviation decreases from 0.0392 to 0.0230, reflecting both improved optimization performance and reduced variability. Because the learner is trained under a time-varying objective induced by adaptive universe selection, stability is nontrivial in this setting. The observed reduction in loss variability under bounded generator drift is qualitatively consistent with the tracking behavior predicted by the theory (Theorem 5.14).

These findings confirm that PRAXIS maintains stable generator dynamics and learner tracking under adaptive universe selection, consistent with the KL-controlled drift framework, and set the stage for analyzing how symbolic archive dynamics further shape robustness.

RQ3: Does the symbolic archive evolve in a structured, selective manner that contributes to robustness beyond adaptive data selection alone?

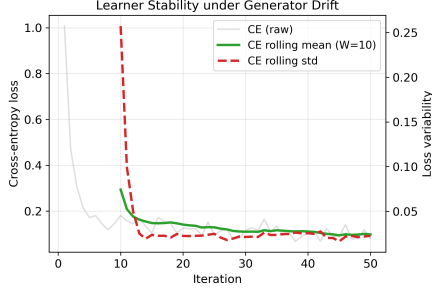


Figure 3. Learner Stability under Non-Stationary Training Distributions

Archive concentration dynamics: Figure 4 shows that the archive distribution becomes increasingly concentrated over training. Archive entropy decreases from 2.28 to 1.96, corresponding to a net reduction of 0.32, while the maximum program weight increases from 0.13 to 0.18. This concentration occurs despite continuous archive expansion via injected symbolic programs, indicating selective reweighting rather than degeneration due to a shrinking hypothesis set. The observed behavior is consistent with the Gibbs-form update $w_{j,t} \propto \exp(\beta s_{j,t})$, under which persistent score differences induce separation in weight space.

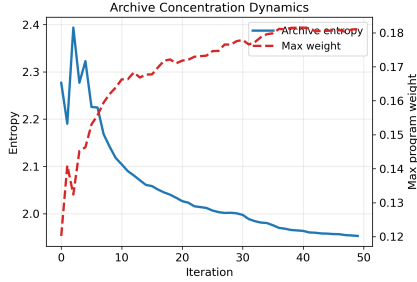


Figure 4. Symbolic Archive Weight Evolution Over Training

Exponential suppression of non-elite programs: Figure 5 analyzes selective suppression by tracking log-weight ratios $\log\left(\frac{w_{j,t}}{\sum_{i \in \mathcal{S}} w_{i,t}}\right)$ for non-elite programs relative to an elite set $\mathcal{S}$ of size $k = 2$. Non-elite programs exhibit approximately linear decay in log scale, corresponding to exponential suppression in weight space. The median non-elite log-ratio decreases with slope -0.489 per iteration, crosses thresholds of -5 , -10 , and -15 at iterations 10, 18, and 28, respectively, and reaches a final value of -25.50 . At this level, cumulative non-elite mass is less than 10^{-11} times the elite mass, rendering these programs effectively inactive in the mixture.

Uniformity and robustness of suppression: Envelope statistics over non-elite programs (Figure 6) show that suppression is not driven by isolated failures. While the worst non-elite program remains near zero, both the median and

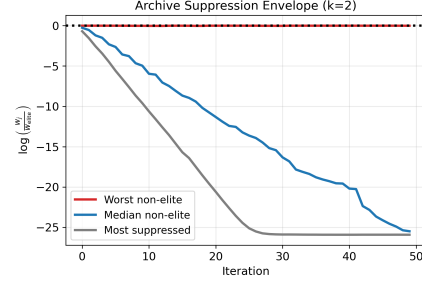


Figure 5. Suppression Envelope of Archive Programs ($k = 2$)

lower envelopes decay steadily toward large negative values. Repeating the analysis for elite sizes $k \in \{1, 2, 3, 5, 10\}$ yields highly consistent behavior: final median log-ratios concentrate within $[-24.8, -25.6]$, and decay slopes remain stable between -0.45 and -0.52 , with threshold-crossing times differing by at most three iterations. Larger elite sets slightly delay early suppression, but do not alter its exponential character or asymptotic scale. This insensitivity to the choice of $\mathcal{S}$ indicates that exponential suppression reflects a structural property of the archive dynamics rather than a tuning artifact.

8. Limitations and Future Work

PRAXIS is evaluated in a controlled setting with a finite universe space and a symbolic archive composed of simple, deterministic programs, which enables precise analysis of KL-controlled generator drift, learner tracking under non-stationarity, and selective archive suppression. This formulation does not yet capture continuous or latent distribution shifts, nor richer symbolic hypothesis classes. Extending the generator to operate over continuous universe parameterizations and the archive to support compositional or probabilistic programs is a natural direction for broadening applicability. From a theoretical standpoint, the current analysis treats generator adaptation, learner optimization, and archive evolution through complementary but modular guarantees; a unified closed-loop treatment of their long-horizon co-evolution remains an important direction for future work.

9. Conclusion

This paper introduced PRAXIS, a unified framework that couples adaptive universe selection, symbolic archive regularization, and gradient-based learning under explicit stability constraints. Theoretical analysis established guarantees on KL-controlled generator drift, learner tracking under non-stationary objectives, and exponential suppression of persistently non-competitive symbolic programs, which were empirically validated through robustness, training stability, and archive evolution analyses.

Impact Statement

PRAXIS has the potential to support safer and more reliable learning in settings subject to distribution shift, such as scientific modeling, decision support, and safety-critical systems. While the framework itself does not introduce direct deployment risks, improper use of adaptive feedback mechanisms may amplify biases or unintended constraints, underscoring the importance of careful design and evaluation when applying such systems in real-world contexts.

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Appendix

A. Related Work

Self-Improving Learning Frameworks. A growing body of work studies self-improving and agentic systems in which models iteratively modify their data, policies, prompts, or intermediate reasoning artifacts through feedback-driven loops, including self-training frameworks (Huang et al., 2025), foundational agent architectures (Xia et al., 2025; Shang et al., 2024), and population-based evolutionary agents (Wang et al., 2025). These approaches demonstrate substantial empirical gains by leveraging archives of prior behaviors, reasoning traces, or candidate solutions, often guided by reward models, verifiers, or task success signals. However, such systems typically emphasize capability growth and system-level behavior, with limited formal analysis of the learning dynamics induced by repeated self-modification, particularly under non-stationary objectives. PRAXIS is complementary to this line of work: rather than proposing a new agent architecture or self-improvement mechanism, it provides a learning-theoretic framework for modeling and analyzing the coupled evolution of data selection, optimization, and symbolic structure that underlies many self-improving systems. In this sense, PRAXIS serves as an abstract dynamical model that captures common mechanisms in agentic and self-evolving frameworks, while enabling explicit guarantees on objective drift, stability, and long-horizon tracking.

Adaptive Data Selection and Learning under Non-Stationary Objectives. ERM (Hanneke & Xu, 2024) constitutes the classical learning paradigm, optimizing average risk under a fixed and stationary data distribution. A large body of work extends ERM by adapting the training distribution over time, thereby implicitly inducing non-stationary learning objectives. Curriculum Learning (CL) methods reorder or reweight examples to accelerate convergence or improve representation learning, beginning with fixed difficulty schedules (Bengio et al., 2009) and extending to modern variants that emphasize training efficiency in large visual backbones (Wang et al., 2023) or integrate active selection criteria (Jafarpour et al., 2021). Distributionally Robust Optimization (DRO) addresses distribution shift by replacing the average-risk objective with a worst-case or group-wise objective, leading to minimax formulations whose solutions evolve as group weights are updated; recent work provides near-optimal algorithms and lower bounds for group DRO (Soma et al., 2025), multi-expert robust formulations (Jeong et al.), and semi-supervised extensions that mitigate group sparsity using unlabeled data (Awasthi et al., 2024; Jung et al.). From a complementary perspective, bandit algorithms formalize adaptive data or policy selection as an online decision problem, balancing exploration and exploitation through regret-minimizing updates or, more recently, enabling valid statistical inference under adaptive data collection (Bibaut et al., 2021; Gutowski et al., 2021). Such methods have also been applied to adaptive components in large language model pipelines, including decoding and tool selection (Bouneffouf & Feraud, 2025). While these approaches differ in mechanism, they share a common structure: the learner is trained against an objective that changes over time as a consequence of adaptive selection, yet the resulting non-stationarity is typically treated as exogenous, adversarial, or heuristic rather than as a first-class dynamical quantity. In contrast, PRAXIS explicitly models data selection, parameter learning, and symbolic structure as a coupled closed-loop system, enabling theoretical guarantees that directly relate the rate and geometry of objective drift to stability and tracking behavior in non-stationary learning environments.

Symbolic and Structured Inductive Bias in Learning. A broad class of neuro-symbolic methods incorporates symbolic structure into learning pipelines to improve compositionality, interpretability, or generalization. Differentiable logic frameworks such as DeepProbLog (Manhaeve et al., 2018) and Scallop (Li et al., 2023) integrate neural predicates with probabilistic or differentiable logic programs, enabling end-to-end training while predominantly assuming a fixed and trusted symbolic program provided as background knowledge. Complementary approaches address non-differentiable symbolic components through gradient estimation techniques, such as Input–Output Sample Gradient Estimation (ISED) (Solko-Breslin et al., 2024), which enables data-efficient optimization of neural programs containing black-box symbolic modules or external APIs without access to internal structure. Other lines of work, including the Neuro-Symbolic Concept Learner (NSCL) (Mao et al., 2019) and related program induction models, learn latent symbolic programs for reasoning over structured representations, demonstrating strong compositional generalization but focusing primarily on task-level accuracy rather than long-horizon learning dynamics. More recent programmable neuro-symbolic systems, such as Dolphin (Naik et al., 2024), combine symbolic modules with adaptive task distributions to scale reasoning across problem families, while symbolic discovery methods extract interpretable rules or expressions from trained neural networks as post-hoc explanations. Across these approaches, symbolic structure is typically treated as fixed, correctness-assumed, or optimized solely for predictive performance, with limited consideration of how symbolic components should be regulated under non-stationary learning conditions. In contrast, PRAXIS treats symbolic programs as fallible inductive biases whose influence is dynamically modulated through interaction with adaptive data selection and parameter learning. By explicitly

modeling the coupled evolution of symbolic structure, data distributions, and model parameters, PRAXIS enables theoretical analysis of stability, suppression, and tracking behavior in learning systems operating under endogenous non-stationarity.

B. More Results

B.1. RQ3: Uniformity and Robustness of Suppression Graphs

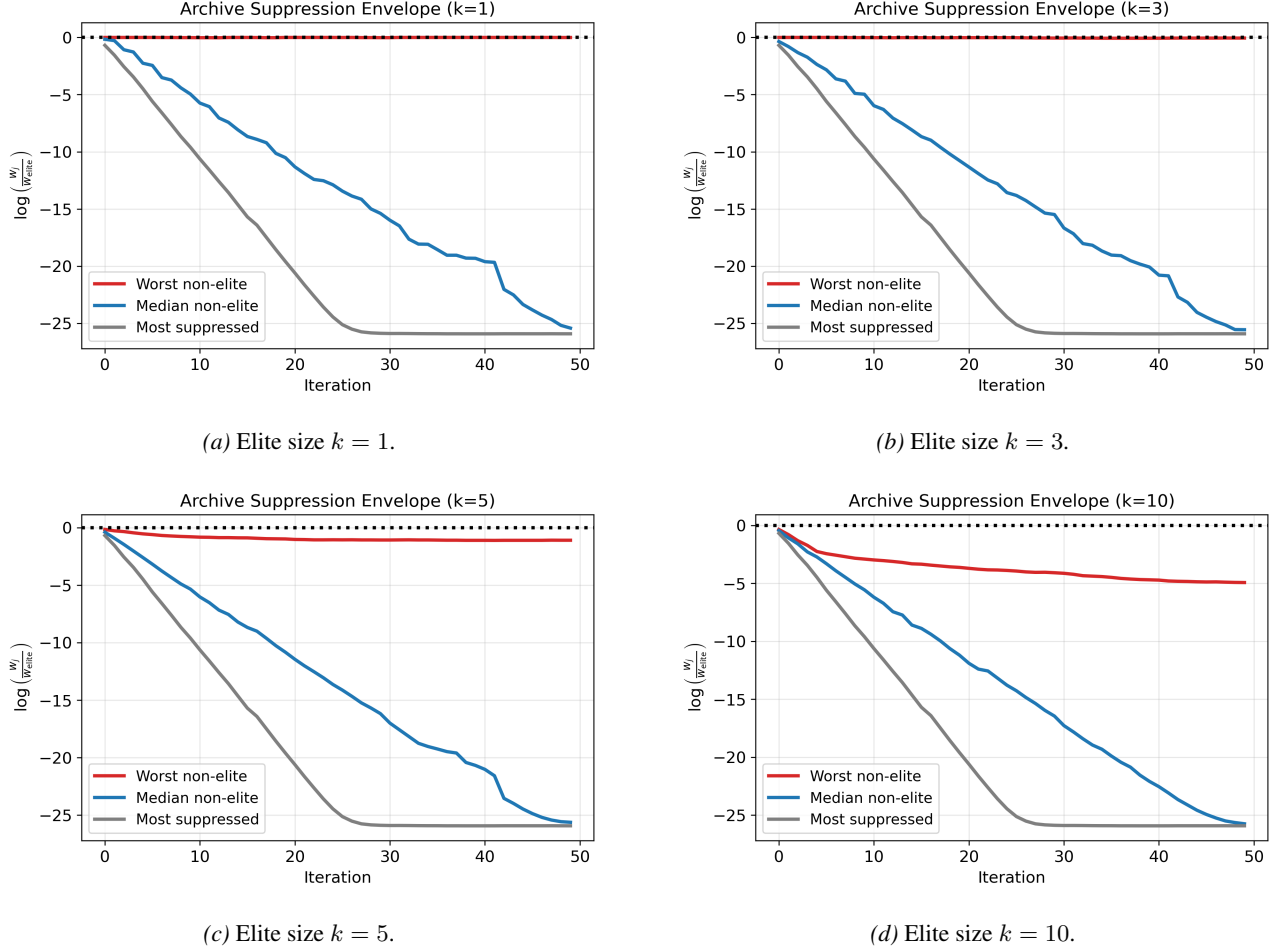


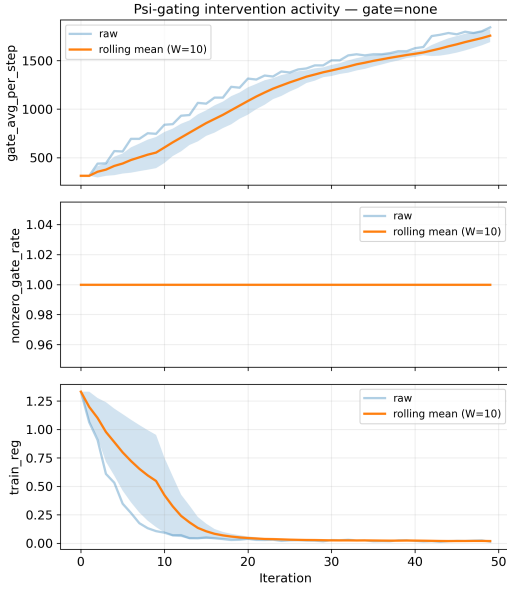
Figure 6. Archive suppression envelopes for additional elite sizes

B.2. Analysis of ψ -Gating Intervention Dynamics

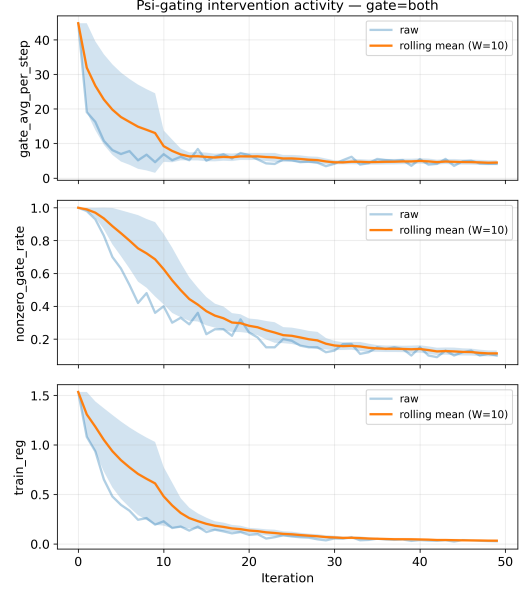
This appendix examines how ψ -gating modulates symbolic intervention frequency and its relationship to robustness and generalization in PRAXIS. We compare four gating policies under fixed thresholds ($\psi_{\text{conf}} = 0.55$, $\psi_{\text{dis}} = 0.40$), chosen a priori to represent moderate confidence and disagreement levels and held constant to isolate the effect of gating structure rather than threshold tuning: no gating (`none`, symbolic interventions applied unconditionally at every step), joint confidence–disagreement gating (`both`), disagreement-only gating (`high-disagree`), and confidence-only gating (`low-conf`). For each configuration, we measure average intervention intensity and the fraction of steps with nonzero intervention, symbolic regularization magnitude, and held-out performance metrics.

Figure 7 shows that ungated training applies symbolic corrections at every step, yielding extremely high intervention pressure (average 1256.5 per step, nonzero rate = 1.0) with limited robustness (held-out $\text{CVaR}_{10} = 0.5703$). In contrast, joint ψ -gating (`both`) reduces intervention frequency by over two orders of magnitude (average 6.68 per step, nonzero rate = 0.281) while achieving the strongest generalization and robustness (held-out $\text{CVaR}_{10} = 0.6406$). Disagreement-only gating triggers frequent interventions (average 316.7 per step, nonzero rate = 1.0) yet underperforms all other configurations,

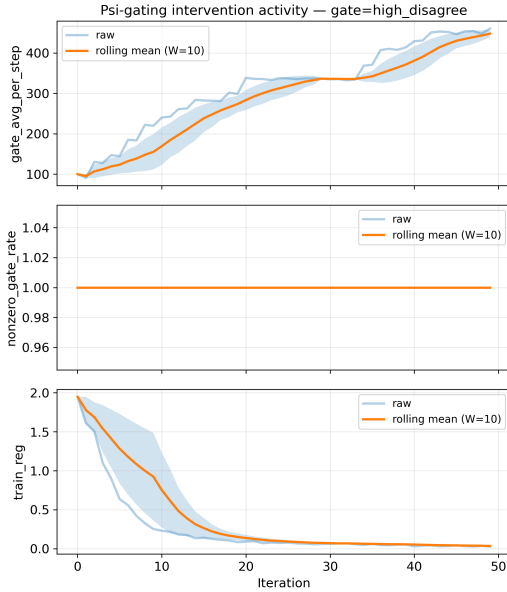
indicating that disagreement alone induces excessive corrective pressure. Confidence-only gating exhibits intermediate behavior (average 11.55 per step, nonzero rate = 0.295) with modest gains over the ungated baseline.



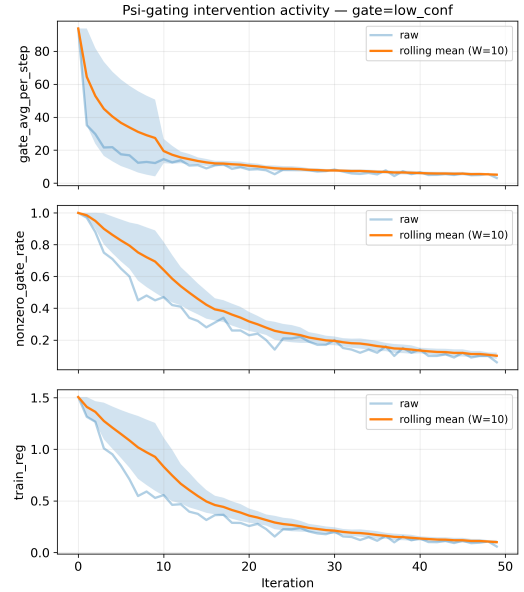
(a) No gating (none)



(b) Confidence + disagreement gating (both)



(c) Disagreement-only gating (high_disagree)



(d) Confidence-only gating (low_conf)

Figure 7. ψ -gating intervention activity over training. Top: average symbolic interventions per step. Middle: fraction of steps with nonzero intervention. Bottom: symbolic regularization magnitude. Selective gating substantially reduces intervention frequency while improving robustness.

Across all gated variants, symbolic regularization decays smoothly over training, indicating that ψ -gating primarily governs the timing and selectivity of symbolic feedback rather than its asymptotic magnitude. Overall, the results demonstrate that sparsifying symbolic interventions through combined confidence and disagreement signals yields more effective and robust learning than either ungated or single-signal strategies.

B.3. Component Ablation Analysis

This subsection evaluates the contribution of PRAXIS components by selectively disabling or freezing adaptive universe selection (generator) and symbolic inductive bias (archive), while keeping all other training and evaluation settings fixed. In addition to fully removing components, partially ablated variants in which either the generator or the archive is held fixed are considered, enabling separation of component availability from adaptive co-evolution effects. The goal is to determine whether robustness and generalization gains arise from individual mechanisms or from their coupled interaction over training.

Table 2. Component ablations of PRAXIS on a fixed test-universe set

Method	Mean	Min	CVaR ₁₀	Std
ERM (fixed mixture, no archive)	0.678	0.617	0.630	0.029
PRAXIS (No archive)	0.658	0.574	0.618	0.038
PRAXIS (Frozen archive)	0.672	0.590	0.624	0.034
PRAXIS (Frozen generator)	0.661	0.589	0.603	0.031
PRAXIS (Full)	0.694	0.602	0.639	0.031

Evaluation follows Table 1. Mean denotes average per-universe accuracy; Min is worst-case accuracy; CVaR₁₀ is the mean accuracy over the bottom 10% universes; Std is the standard deviation across test universes.

Disabling the symbolic archive while retaining adaptive universe selection degrades robustness across all tail metrics: CVaR₁₀ decreases from 0.639 (full PRAXIS) to 0.618, worst-case accuracy drops from 0.602 to 0.574, and variability across universes increases (Std 0.038 vs. 0.031). This indicates that adaptive data selection alone is insufficient to preserve worst-case performance without symbolic inductive bias. Freezing the symbolic archive partially recovers performance relative to the no-archive variant (Mean 0.672 vs. 0.658, CVaR₁₀ 0.624 vs. 0.618), but remains consistently below full PRAXIS across all robustness metrics. This gap demonstrates that the benefits of the archive arise not merely from the presence of symbolic rules, but from their adaptive reweighting in response to shifting data distributions. Conversely, freezing the generator while retaining an adaptive archive yields improved stability relative to the no-archive condition (Std 0.031 vs. 0.038) but further reduces tail robustness (CVaR₁₀ = 0.603), indicating that symbolic structure alone cannot compensate for the absence of adaptive universe selection. Full PRAXIS achieves the strongest average performance (Mean 0.694) and tail robustness (CVaR₁₀ = 0.639) while maintaining low variance, indicating that robustness gains arise from the coupled interaction between controlled data drift and adaptive symbolic regularization, rather than from either component in isolation.

C. Dataset Details

Experiments are conducted on a controlled subset of the MIT Indoor Scenes dataset, a standard benchmark for scene recognition that contains diverse indoor environments with substantial visual variability. To enable precise analysis of learning dynamics under adaptive data selection, a fixed subset of 12 scene classes is used, with an equal number of images per class. Specifically, 100 images are selected from each class, yielding a balanced dataset of 1,200 images in total. The selected classes span a range of spatial layouts, object densities, and semantic contexts, including residential, commercial, and public indoor spaces. This diversity allows robustness and generalization effects to be measured across meaningfully different visual environments while maintaining tight control over class balance and dataset scale. The use of a fixed, balanced subset serves two purposes. First, it avoids confounding effects due to class imbalance or long-tailed distributions, ensuring that observed performance differences arise from adaptive training dynamics rather than data frequency artifacts. Second, the moderate dataset size enables repeated evaluation across many synthetic “universes” with varying augmentation and corruption parameters, which is central to the PRAXIS experimental design.

Table 3. Indoor scene classes used from the MIT Indoor dataset

auditorium	bedroom	bookstore	classroom	computer room	corridor
grocerystore	kitchen	library	living room	office	restaurant

D. Implementation and Training Details

D.1. Training Universe Construction

Training environments are modeled as a finite discrete universe space $\mathcal{U}$. Each universe corresponds to a specific configuration of:

- Training set size: $\{400, 800, 1200, 1600, 2000\}$,
- Augmentation strength: 5 values uniformly spaced in $[0, 0.5]$,
- Corruption type: $\{\text{none}, \text{blur}, \text{jpeg}, \text{noise}\}$,
- Corruption severity: $\{0.0, 0.3, 0.6\}$.

The Cartesian product of these parameters yields $5 \times 5 \times 4 \times 3 = 300$ distinct training universes. All adaptive methods, including PRAXIS, curriculum learning, bandit sampling, and Group DRO, operate over this same universe set during training. Here, training set size refers to the number of samples drawn with replacement per iteration from the underlying 1,200-image dataset, using on-the-fly augmentations and corruptions. Thus, reported sizes correspond to the number of stochastic draws rather than distinct images.

D.2. Held-Out Evaluation Universes

Robustness is evaluated on a separate set of held-out universes that are disjoint from the training universe grid. Held-out universes are constructed using:

- Evaluation set size: $\{2400\}$,
- Augmentation strengths: $\{0.5, 0.75, 0.9\}$,
- Corruption types: $\{\text{blur}, \text{jpeg}, \text{noise}\}$,
- Corruption severities: $\{0.5, 0.75, 0.85\}$.

The held-out evaluation set size denotes the number of augmented samples drawn with replacement from a fixed validation split. Held-out universes are never sampled during training and are used exclusively for evaluation. The evaluation protocol is designed to induce a controlled distribution shift rather than a domain shift.

E. Composition of the Symbolic Archive

The symbolic archive in PRAXIS is not intended to encode domain knowledge or task-specific rules. Instead, it serves as a structured hypothesis space over the geometry of the learner’s predictive distribution. Each archive program deterministically maps the model’s class-probability vector to a hard label, enabling the archive to impose selective, interpretable regularization on the learner without accessing raw inputs, gradients, or labels. The initial archive is manually seeded with a small set of programs designed to span qualitatively distinct families of structural inductive bias. These families are chosen to probe different failure modes and uncertainty regimes of modern neural classifiers:

- **Ranking-based rules** (e.g., argmax, second-best, top-2 margin switching) capture sensitivity to near-ties and decision boundary ambiguity.
- **Threshold-based rules** (e.g., confidence and entropy gating) explicitly react to predictive uncertainty, separating confident predictions from diffuse or high-entropy outputs.
- **Symmetry-breaking biases** (e.g., even/odd preference, low-index bias) introduce controlled asymmetries that test whether the learner’s confidence is robust to benign structural perturbations.
- **Temporal and modular patterns** (e.g., modulo-based selection conditioned on timestep) inject simple non-stationary structure, allowing the archive to express time-varying preferences without direct access to data.

- **Distributional preferences** (e.g., center-biased selection) encode priors over label geometry rather than semantic content.

Importantly, all programs operate on the the learner’s predictive distribution and differ only in how that distribution is interrogated. The archive programs are intentionally lightweight, but not simplistic. They are expressive enough to induce meaningful disagreement with the learner under distribution shift, yet constrained enough to avoid encoding task-specific shortcuts. This balance is critical: overly expressive programs would collapse into surrogate models, while overly trivial rules would be systematically suppressed by the fitness–blame mechanism.

Program proposals are generated using the GPT-4o language model through a structured prompting interface that restricts outputs to a fixed library of predefined symbolic templates with bounded parameters. The model is instructed to return a small set of candidate programs in a strict JSON format, specifying only a template identifier, parameter values, and optional metadata. The prompt structure used in the implementation is shown below.

Listing 1. JSON-only prompt used for template-constrained program proposals.

```
{
  "task": "Propose a small set of symbolic programs (hard-label rules) to add to a
  PRAXIS archive.",
  "constraints": [
    "You must ONLY use the provided templates; do NOT invent new templates.",
    "Return STRICT JSON only, no markdown, no commentary.",
    "Return an object with key 'programs' mapping to a list of program specs.",
    "Each program spec must have: template (string), params (object), name (string
    , optional), family='gpt'.",
    "Keep it diverse: propose programs that differ in template and parameters.",
    "Suggest at most N programs as requested.",
    f"Classes are indexed 0..{num_classes-1}."
  ],
  "available_templates": templates_desc,
  "current_context": context,
  "output_schema": {
    "programs": [
      {
        "template": "parity_bias", "params": {"parity": 0}, "name": "optional", "
        family": "gpt"
      }
    ]
  },
}
```

This design ensures that the LLM contributes diversity without introducing opaque decision-making or bypassing the formal learning dynamics of PRAXIS. The LLM cannot inject new symbolic primitives, alter archive update rules, or encode task-specific knowledge beyond what is already captured by the predefined templates and contextual metadata. As a result, any long-term influence of LLM-proposed programs on learning behavior arises only if those programs are empirically useful under the same stability-controlled archive evolution analyzed in the main text.

F. Theory Proofs

F.1. Proofs for Section 5.1

Proof of Theorem 5.4. Fix any $t \geq 0$ and $\theta \in \Theta$. By Definition 5.1,

$$F_{t+1}(\theta) - F_t(\theta) = \mathbb{E}_{u \sim q_{t+1}}[L_{\text{aug}}(\theta, u)] - \mathbb{E}_{u \sim q_t}[L_{\text{aug}}(\theta, u)]. \quad (6)$$

Since $q_t, q_{t+1} \in \Delta(\mathcal{U})$ are probability measures on $\mathcal{U}$, this difference can be written as

$$F_{t+1}(\theta) - F_t(\theta) = \int_{\mathcal{U}} L_{\text{aug}}(\theta, u) (q_{t+1}(u) - q_t(u)) du. \quad (7)$$

Taking absolute values and applying the triangle inequality yields

$$|F_{t+1}(\theta) - F_t(\theta)| \leq \int_{\mathcal{U}} |L_{\text{aug}}(\theta, u)| |q_{t+1}(u) - q_t(u)| du. \quad (8)$$

Using the uniform bound $\sup_{\theta \in \Theta, u \in \mathcal{U}} |L_{\text{aug}}(\theta, u)| \leq M$, we obtain

$$|F_{t+1}(\theta) - F_t(\theta)| \leq M \int_{\mathcal{U}} |q_{t+1}(u) - q_t(u)| du = M \|q_{t+1} - q_t\|_1. \quad (9)$$

Because this bound holds for every $\theta \in \Theta$, taking the supremum over Θ gives

$$\delta_t = \sup_{\theta \in \Theta} |F_{t+1}(\theta) - F_t(\theta)| \leq M \|q_{t+1} - q_t\|_1. \quad (10)$$

We next relate the ℓ_1 distance to the KL divergence. If $\text{KL}(q_{t+1} \| q_t) < \infty$, then q_{t+1} is absolutely continuous with respect to q_t , and Pinsker's inequality applies. Specifically, Pinsker's inequality states that

$$\|q_{t+1} - q_t\|_1 \leq \sqrt{2 \text{KL}(q_{t+1} \| q_t)}. \quad (11)$$

Combining the above inequalities yields

$$\delta_t \leq M \|q_{t+1} - q_t\|_1 \leq M \sqrt{2 \text{KL}(q_{t+1} \| q_t)}, \quad (12)$$

establishing the claimed KL-controlled drift bound.

Finally, suppose that the Generator sequence satisfies $\sum_{t=0}^{\infty} \text{KL}(q_{t+1} \| q_t) < \infty$. Since each KL term is nonnegative, summability implies $\text{KL}(q_{t+1} \| q_t) \rightarrow 0$ as $t \rightarrow \infty$. Substituting into the above bound gives

$$\delta_t \leq M \sqrt{2 \text{KL}(q_{t+1} \| q_t)} \rightarrow 0, \quad \text{as } t \rightarrow \infty.$$

Hence, proved. $\square$

Proof of Lemma 5.6. Fix any $t \geq 0$. For every realization of the algorithmic randomness up to iteration $t + 1$, we have $\theta_{t+1} \in \Theta$. Hence,

$$|F_{t+1}(\theta_{t+1}) - F_t(\theta_{t+1})| \leq \sup_{\theta \in \Theta} |F_{t+1}(\theta) - F_t(\theta)| = \delta_t.$$

Taking expectations over the randomness preserves the inequality, yielding $\Delta_t \leq \delta_t$. $\square$

Proof of Corollary 5.8. Fix any $t \geq T_0$. By the first inequality in Theorem 5.4,

$$\delta_t = \sup_{\theta \in \Theta} |F_{t+1}(\theta) - F_t(\theta)| \leq M \|q_{t+1} - q_t\|_1. \quad (13)$$

By exponential Generator stabilization (Definition 5.7), $\|q_{t+1} - q_t\|_1 \leq c e^{-\kappa(t-T_0)}$. Substituting yields

$$\delta_t \leq M c e^{-\kappa(t-T_0)}.$$

Finally, Lemma 5.6 gives $\Delta_t \leq \delta_t$, and therefore

$$\Delta_t \leq \delta_t \leq M c e^{-\kappa(t-T_0)}, \quad \forall t \geq T_0.$$

Hence, proved. $\square$

F.2. Proofs for Section 5.2

Proof of Lemma 5.11. Fix an arbitrary iteration t and program index k .

Step 1: Bounding the blame term. By assumption, blame is learner-gated i.e., blame is incurred only on examples that the learner classifies correctly at iteration t . Equivalently, $B_{k,t}$ is an expectation of a bounded indicator (or bounded loss) multiplied by the learner-correctness gate $\mathbb{1}\{\hat{y}_t(x) = y\}$. In particular, for every x , the per-sample blame contribution is nonnegative and at most 1 whenever the gate is active, and it is 0 when the gate is inactive. Therefore, almost surely,

$$0 \leq (\text{per-sample blame contribution}) \leq \mathbb{1}\{\hat{y}_t(x) = y\}.$$

Taking expectations over (x, y) yields

$$0 \leq B_{k,t} \leq \mathbb{P}(\hat{y}_t(x) = y).$$

By definition of the learner population error $e_t := \mathbb{P}(\hat{y}_t(x) \neq y)$, we have $\mathbb{P}(\hat{y}_t(x) = y) = 1 - e_t$, hence

$$0 \leq B_{k,t} \leq 1 - e_t. \quad (14)$$

Step 2: Lower bounding utility gaps. Recall the population utility definition $U_{j,t} := F_{j,t} - \gamma B_{j,t}$ with $\gamma \geq 0$. For any two programs i and j ,

$$\begin{aligned} U_{i,t} - U_{j,t} &= (F_{i,t} - \gamma B_{i,t}) - (F_{j,t} - \gamma B_{j,t}) \\ &= (F_{i,t} - F_{j,t}) - \gamma(B_{i,t} - B_{j,t}). \end{aligned} \quad (15)$$

Using the blame bounds (14) for each program, we have $0 \leq B_{i,t} \leq 1 - e_t$ and $0 \leq B_{j,t} \leq 1 - e_t$, which implies

$$B_{i,t} - B_{j,t} \leq B_{i,t} \leq 1 - e_t.$$

Substituting into (15) gives

$$U_{i,t} - U_{j,t} \geq (F_{i,t} - F_{j,t}) - \gamma(1 - e_t).$$

Define

$$\varepsilon_{i,j,t}^{\text{endo}} := (F_{i,t} - F_{j,t}) - \gamma(1 - e_t).$$

Then the preceding inequality is exactly $U_{i,t} - U_{j,t} \geq \varepsilon_{i,j,t}^{\text{endo}}$, as claimed. Hence, proved. $\square$

Proof of Theorem 5.12. Fix $T \geq T_0$ and $\delta \in (0, 1)$. Unroll the score recursion:

$$s_{j,t} = s_{j,T_0} + \eta_p \sum_{\tau=T_0}^{t-1} \hat{U}_{j,\tau}, \quad \hat{U}_{j,\tau} = U_{j,\tau} + \xi_{j,\tau}.$$

Step 1: Reduce the ratio to score differences. For any t and any j ,

$$\frac{w_{j,t}}{\sum_{i \in \mathcal{S}} w_{i,t}} = \frac{e^{\beta s_{j,t}}}{\sum_{i \in \mathcal{S}} e^{\beta s_{i,t}}}.$$

Since $\sum_{i \in \mathcal{S}} e^{\beta s_{i,t}} \geq e^{\beta s_{i,t}}$ for every fixed $i \in \mathcal{S}$, we have for all $i \in \mathcal{S}$,

$$\frac{w_{j,t}}{\sum_{i' \in \mathcal{S}} w_{i',t}} \leq \exp(\beta(s_{j,t} - s_{i,t})). \quad (16)$$

Likewise, using $\sum_{i' \in \mathcal{S}} w_{i',T_0} \geq w_{i,T_0}$,

$$\exp(\beta(s_{j,T_0} - s_{i,T_0})) = \frac{w_{j,T_0}}{w_{i,T_0}} \leq \frac{w_{j,T_0}}{\sum_{i' \in \mathcal{S}} w_{i',T_0}}. \quad (17)$$

Step 2: Bound score differences by cumulative gaps and cumulative noise. Fix $j \notin \mathcal{S}$, $t \in \{T_0, \dots, T\}$, and any $i \in \mathcal{S}$. Unrolling gives

$$s_{j,t} - s_{i,t} = (s_{j,T_0} - s_{i,T_0}) + \eta_p \sum_{\tau=T_0}^{t-1} (\hat{U}_{j,\tau} - \hat{U}_{i,\tau}).$$

Substitute $\hat{U}_{k,\tau} = U_{k,\tau} + \xi_{k,\tau}$:

$$\sum_{\tau=T_0}^{t-1} (\hat{U}_{j,\tau} - \hat{U}_{i,\tau}) = \sum_{\tau=T_0}^{t-1} (U_{j,\tau} - U_{i,\tau}) + \sum_{\tau=T_0}^{t-1} (\xi_{j,\tau} - \xi_{i,\tau}).$$

By the definition of $\varepsilon_{j,\tau}^{\mathcal{S}}$, for every τ and every $i \in \mathcal{S}$,

$$U_{i,\tau} - U_{j,\tau} \leq \max_{i' \in \mathcal{S}} (U_{i',\tau} - U_{j,\tau}) = \varepsilon_{j,\tau}^{\mathcal{S}},$$

hence

$$U_{j,\tau} - U_{i,\tau} \leq -\varepsilon_{j,\tau}^{\mathcal{S}} \leq -\underline{\varepsilon}_{\tau}^{\mathcal{S}}.$$

Summing over $\tau = T_0, \dots, t-1$ yields

$$\sum_{\tau=T_0}^{t-1} (U_{j,\tau} - U_{i,\tau}) \leq - \sum_{\tau=T_0}^{t-1} \underline{\varepsilon}_{\tau}^{\mathcal{S}}. \quad (18)$$

Therefore,

$$s_{j,t} - s_{i,t} \leq (s_{j,T_0} - s_{i,T_0}) - \eta_p \sum_{\tau=T_0}^{t-1} \underline{\varepsilon}_{\tau}^{\mathcal{S}} + \eta_p \sum_{\tau=T_0}^{t-1} (\xi_{j,\tau} - \xi_{i,\tau}). \quad (19)$$

Step 3: Uniform high-probability bound on cumulative noise differences. Define conditional means $m_{k,\tau} := \mathbb{E}[\xi_{k,\tau} \mid \mathcal{F}_{\tau-1}]$. By Definition 5.10(i), $|m_{k,\tau}| \leq \mu$ a.s., so for any i, j ,

$$\sum_{\tau=T_0}^{t-1} (m_{j,\tau} - m_{i,\tau}) \leq \sum_{\tau=T_0}^{t-1} (|m_{j,\tau}| + |m_{i,\tau}|) \leq 2\mu(t - T_0). \quad (20)$$

Let the centered pairwise differences be

$$d_{\tau}^{(i,j)} := (\xi_{j,\tau} - \xi_{i,\tau}) - \mathbb{E}[\xi_{j,\tau} - \xi_{i,\tau} \mid \mathcal{F}_{\tau-1}].$$

Then $\mathbb{E}[d_{\tau}^{(i,j)} \mid \mathcal{F}_{\tau-1}] = 0$ and by Definition 5.10(ii), for all $\lambda \in \mathbb{R}$,

$$\mathbb{E}\left[e^{\lambda d_{\tau}^{(i,j)}} \mid \mathcal{F}_{\tau-1}\right] \leq \exp\left(\frac{\lambda^2 (2\sigma)^2}{2}\right) \quad \text{a.s.}$$

Let $M_t^{(i,j)} := \sum_{\tau=T_0}^{t-1} d_{\tau}^{(i,j)}$. A standard Chernoff argument using the conditional MGF bound yields, for any $a > 0$,

$$\mathbb{P}\left(M_t^{(i,j)} \geq a\right) \leq \exp\left(-\frac{a^2}{2(2\sigma)^2(t - T_0)}\right) = \exp\left(-\frac{a^2}{8\sigma^2(t - T_0)}\right). \quad (21)$$

Apply a union bound over all ordered pairs $(i, j) \in [m] \times [m]$ and all $t \in \{T_0, \dots, T\}$, for a total of $m^2(T - T_0 + 1)$ events. Setting

$$a(t, \delta) = 2\sigma \sqrt{2(t - T_0) \log\left(\frac{m^2(T - T_0 + 1)}{\delta}\right)},$$

and using (21), we obtain that with probability at least $1 - \delta$, simultaneously for all (i, j) and all $t \in \{T_0, \dots, T\}$,

$$\sum_{\tau=T_0}^{t-1} (\xi_{j,\tau} - \xi_{i,\tau}) \leq 2\mu(t - T_0) + 2\sigma \sqrt{2(t - T_0) \log\left(\frac{m^2(T - T_0 + 1)}{\delta}\right)}. \quad (22)$$

Step 4: Conclude the suppression bound. On the event in (22), substitute into (19) to obtain, for all $i \in \mathcal{S}$,

$$s_{j,t} - s_{i,t} \leq (s_{j,T_0} - s_{i,T_0}) - \eta_p \sum_{\tau=T_0}^{t-1} \underline{\varepsilon}_{\tau}^{\mathcal{S}} + \eta_p \Gamma_{\text{subG}}(t, \delta),$$

where $\Gamma_{\text{subG}}(t, \delta)$ is as in the theorem. Multiply by β and exponentiate:

$$\exp(\beta(s_{j,t} - s_{i,t})) \leq \exp(\beta(s_{j,T_0} - s_{i,T_0})) \exp\left(-\beta\eta_p \sum_{\tau=T_0}^{t-1} \underline{\varepsilon}_{\tau}^{\mathcal{S}} + \beta\eta_p \Gamma_{\text{subG}}(t, \delta)\right).$$

Finally apply (16) and (17):

$$\frac{w_{j,t}}{\sum_{i' \in \mathcal{S}} w_{i',t}} \leq \exp(\beta(s_{j,t} - s_{i,t})) \leq \frac{w_{j,T_0}}{\sum_{i' \in \mathcal{S}} w_{i',T_0}} \exp\left(-\beta\eta_p \sum_{\tau=T_0}^{t-1} \underline{\varepsilon}_{\tau}^{\mathcal{S}} + \beta\eta_p \Gamma_{\text{subG}}(t, \delta)\right),$$

which holds uniformly for all $t \in \{T_0, \dots, T\}$ and all $j \notin \mathcal{S}$ on an event of probability at least $1 - \delta$. $\square$

Proof of Corollary 5.13. Fix a horizon $T \geq T_0$ and confidence level $\delta \in (0, 1)$. By Theorem 5.12, with probability at least $1 - \delta$, the following holds simultaneously for all $t \in \{T_0, \dots, T\}$ and all $j \notin \mathcal{S}$:

$$\frac{w_{j,t}}{\sum_{i \in \mathcal{S}} w_{i,t}} \leq \frac{w_{j,T_0}}{\sum_{i \in \mathcal{S}} w_{i,T_0}} \exp\left(-\beta\eta_p \sum_{\tau=T_0}^{t-1} \underline{\varepsilon}_\tau^{\mathcal{S}} + \beta\eta_p \Gamma_{\text{subG}}(t, \delta)\right). \quad (23)$$

Taking logarithms of both sides of (23) yields

$$\log \frac{w_{j,t}}{\sum_{i \in \mathcal{S}} w_{i,t}} \leq \log \frac{w_{j,T_0}}{\sum_{i \in \mathcal{S}} w_{i,T_0}} - \beta\eta_p \sum_{\tau=T_0}^{t-1} \underline{\varepsilon}_\tau^{\mathcal{S}} + \beta\eta_p \Gamma_{\text{subG}}(t, \delta). \quad (24)$$

Now fix $t \in \{T_1, \dots, T\}$. Since $\max(\underline{\varepsilon}_\tau^{\mathcal{S}}, 0) \geq \underline{\varepsilon}_\tau^{\mathcal{S}}$ for every τ , we have

$$\sum_{\tau=T_0}^{t-1} \underline{\varepsilon}_\tau^{\mathcal{S}} \geq \sum_{\tau=T_1}^{t-1} \underline{\varepsilon}_\tau^{\mathcal{S}} \geq \sum_{\tau=T_1}^{t-1} \max(\underline{\varepsilon}_\tau^{\mathcal{S}}, 0).$$

By the persistent positive margin assumption in the statement of the corollary,

$$\sum_{\tau=T_1}^{t-1} \max(\underline{\varepsilon}_\tau^{\mathcal{S}}, 0) \geq \varepsilon(t - T_1).$$

Combining the last two displays gives

$$\sum_{\tau=T_0}^{t-1} \underline{\varepsilon}_\tau^{\mathcal{S}} \geq \varepsilon(t - T_1), \quad \forall t \in \{T_1, \dots, T\}. \quad (25)$$

Substituting (25) into (24) yields

$$\log \frac{w_{j,t}}{\sum_{i \in \mathcal{S}} w_{i,t}} \leq \underbrace{\log \frac{w_{j,T_0}}{\sum_{i \in \mathcal{S}} w_{i,T_0}} + \beta\eta_p \sum_{\tau=T_0}^{T_1-1} |\underline{\varepsilon}_\tau^{\mathcal{S}}|}_{=: C} - \beta\eta_p \varepsilon(t - T_1) + \beta\eta_p \Gamma_{\text{subG}}(t, \delta), \quad j \notin \mathcal{S},$$

where C is a finite constant that depends only on the initial weights (and the finite prefix up to T_1), hence depends only on the state at time T_1 as claimed.

Finally, if $\mu = 0$ then $\Gamma_{\text{subG}}(t, \delta) = 2\sigma\sqrt{2(t - T_0) \log(\frac{m(T - T_0 + 1)}{\delta})}$ grows on the order of $\sqrt{t}$, whereas the deterministic term $\varepsilon(t - T_1)$ grows linearly in t . Thus the exponent is eventually dominated by the linear decay term, yielding exponential suppression up to sublinear stochastic fluctuations. $\square$

F.3. Proof for Section 5.3

Proof of Theorem 5.14. Fix $t \geq 0$. By L -smoothness of F_t (Descent Lemma), for any $\theta, \theta' \in \Theta$,

$$F_t(\theta') \leq F_t(\theta) + \langle \nabla F_t(\theta), \theta' - \theta \rangle + \frac{L}{2} \|\theta' - \theta\|^2.$$

Apply this with $\theta = \theta_t$ and $\theta' = \theta_{t+1} = \theta_t - \eta g_t$ to obtain

$$\begin{aligned} F_t(\theta_{t+1}) &\leq F_t(\theta_t) + \langle \nabla F_t(\theta_t), -\eta g_t \rangle + \frac{L}{2} \|\eta g_t\|^2 \\ &= F_t(\theta_t) - \eta \langle \nabla F_t(\theta_t), g_t \rangle + \frac{L\eta^2}{2} \|g_t\|^2. \end{aligned} \quad (26)$$

Now take conditional expectation of (26) w.r.t. $\mathcal{F}_{t-1}$. Since θ_t is $\mathcal{F}_{t-1}$ -measurable and F_t is fixed given $\mathcal{F}_{t-1}$,

$$\mathbb{E}[F_t(\theta_{t+1}) \mid \mathcal{F}_{t-1}] \leq F_t(\theta_t) - \eta \mathbb{E}[\langle \nabla F_t(\theta_t), g_t \rangle \mid \mathcal{F}_{t-1}] + \frac{L\eta^2}{2} \mathbb{E}[\|g_t\|^2 \mid \mathcal{F}_{t-1}]. \quad (27)$$

Step 1: Unbiasedness term. By assumption (i), $\mathbb{E}[g_t \mid \mathcal{F}_{t-1}] = \nabla F_t(\theta_t)$. Therefore,

$$\mathbb{E}[\langle \nabla F_t(\theta_t), g_t \rangle \mid \mathcal{F}_{t-1}] = \langle \nabla F_t(\theta_t), \mathbb{E}[g_t \mid \mathcal{F}_{t-1}] \rangle = \|\nabla F_t(\theta_t)\|^2.$$

Substitute into (27).

Step 2: Second-moment term. Decompose $g_t = \nabla F_t(\theta_t) + (g_t - \nabla F_t(\theta_t))$ and expand:

$$\begin{aligned} \mathbb{E}[\|g_t\|^2 \mid \mathcal{F}_{t-1}] &= \mathbb{E}[\|\nabla F_t(\theta_t)\|^2 + 2 \langle \nabla F_t(\theta_t), g_t - \nabla F_t(\theta_t) \rangle + \|g_t - \nabla F_t(\theta_t)\|^2 \mid \mathcal{F}_{t-1}] \\ &= \|\nabla F_t(\theta_t)\|^2 + \mathbb{E}[\|g_t - \nabla F_t(\theta_t)\|^2 \mid \mathcal{F}_{t-1}], \end{aligned}$$

where the cross term vanishes because $\mathbb{E}[g_t - \nabla F_t(\theta_t) \mid \mathcal{F}_{t-1}] = 0$ by unbiasedness. Using assumption (ii),

$$\mathbb{E}[\|g_t\|^2 \mid \mathcal{F}_{t-1}] \leq \|\nabla F_t(\theta_t)\|^2 + \sigma^2.$$

Combining the two displayed bounds with (27) yields

$$\begin{aligned} \mathbb{E}[F_t(\theta_{t+1}) \mid \mathcal{F}_{t-1}] &\leq F_t(\theta_t) - \eta \|\nabla F_t(\theta_t)\|^2 + \frac{L\eta^2}{2} (\|\nabla F_t(\theta_t)\|^2 + \sigma^2) \\ &= F_t(\theta_t) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla F_t(\theta_t)\|^2 + \frac{L\eta^2}{2} \sigma^2. \end{aligned} \quad (28)$$

Assume $\eta \in (0, 1/L]$. Then $1 - \frac{L\eta}{2} \geq \frac{1}{2}$, and (28) implies

$$\mathbb{E}[F_t(\theta_{t+1}) \mid \mathcal{F}_{t-1}] \leq F_t(\theta_t) - \frac{\eta}{2} \|\nabla F_t(\theta_t)\|^2 + \frac{L\eta^2}{2} \sigma^2. \quad (29)$$

Step 3: Inject objective drift. By definition of the uniform drift, for any $\theta \in \Theta$,

$$F_{t+1}(\theta) \leq F_t(\theta) + \delta_t.$$

In particular, this holds at $\theta = \theta_{t+1}$, giving

$$F_{t+1}(\theta_{t+1}) \leq F_t(\theta_{t+1}) + \delta_t. \quad (30)$$

Taking conditional expectation of (30) given $\mathcal{F}_{t-1}$ and combining with (29) yields

$$\mathbb{E}[F_{t+1}(\theta_{t+1}) \mid \mathcal{F}_{t-1}] \leq F_t(\theta_t) - \frac{\eta}{2} \|\nabla F_t(\theta_t)\|^2 + \frac{L\eta^2}{2} \sigma^2 + \delta_t.$$

Now take total expectation to obtain

$$\mathbb{E}[F_{t+1}(\theta_{t+1})] \leq \mathbb{E}[F_t(\theta_t)] - \frac{\eta}{2} \mathbb{E}[\|\nabla F_t(\theta_t)\|^2] + \frac{L\eta^2}{2} \sigma^2 + \delta_t. \quad (31)$$

Step 4: Sum and telescope. Sum (31) over $t = 0, 1, \dots, T-1$:

$$\mathbb{E}[F_T(\theta_T)] \leq \mathbb{E}[F_0(\theta_0)] - \frac{\eta}{2} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla F_t(\theta_t)\|^2] + \frac{L\eta^2}{2} T \sigma^2 + \sum_{t=0}^{T-1} \delta_t.$$

Rearranging gives

$$\frac{\eta}{2} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla F_t(\theta_t)\|^2] \leq \mathbb{E}[F_0(\theta_0)] - \mathbb{E}[F_T(\theta_T)] + \frac{L\eta^2}{2} T\sigma^2 + \sum_{t=0}^{T-1} \delta_t.$$

Using the lower bound $F_T(\theta_T) \geq F_{\min}$ (and hence $\mathbb{E}[F_T(\theta_T)] \geq F_{\min}$),

$$\frac{\eta}{2} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla F_t(\theta_t)\|^2] \leq F_0(\theta_0) - F_{\min} + \frac{L\eta^2}{2} T\sigma^2 + \sum_{t=0}^{T-1} \delta_t.$$

Divide both sides by T and multiply by $2/\eta$ to obtain

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla F_t(\theta_t)\|^2] \leq \frac{2}{\eta T} (F_0(\theta_0) - F_{\min}) + \eta L \sigma^2 + \frac{2}{T} \sum_{t=0}^{T-1} \delta_t,$$

which is exactly the claim. Hence, proved. □